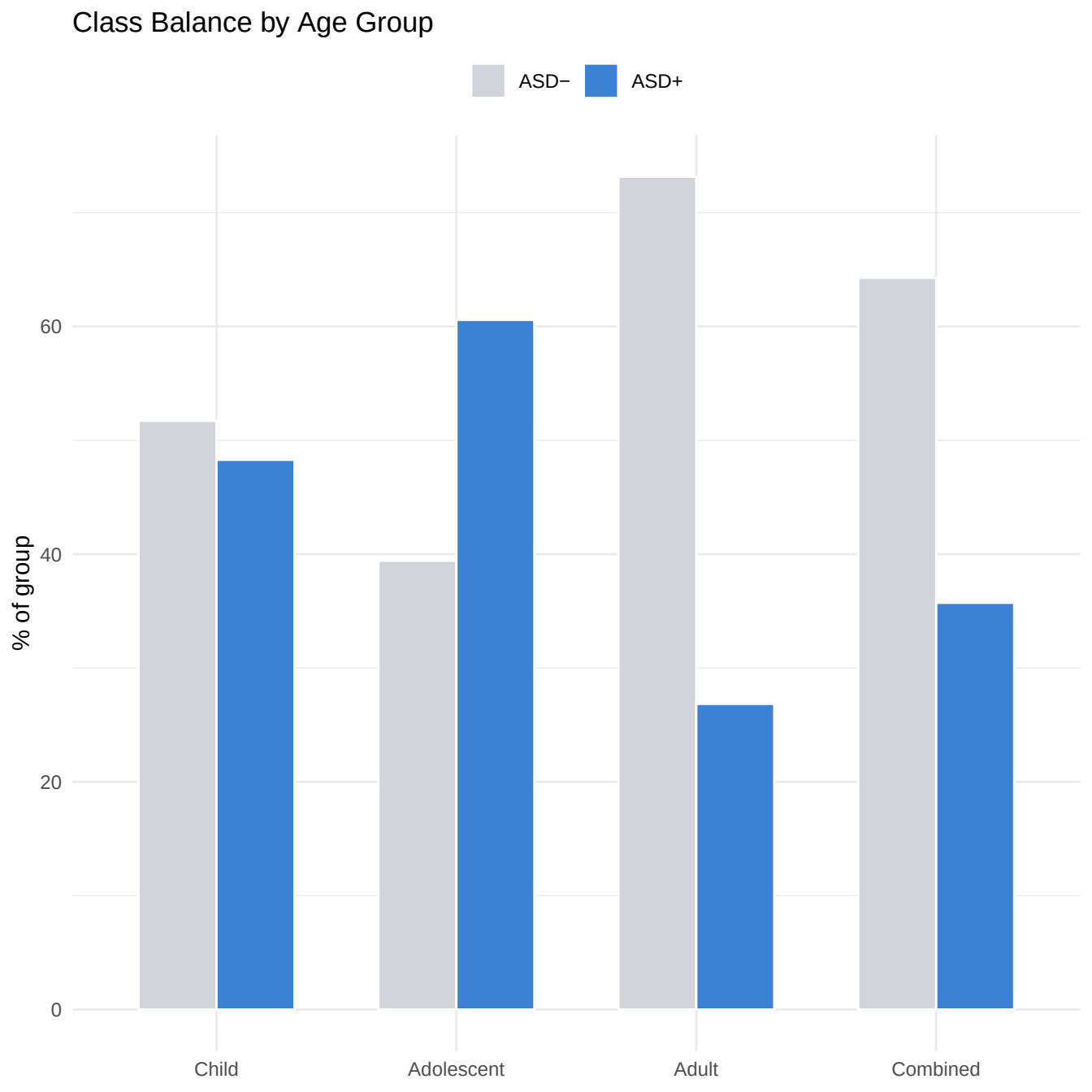
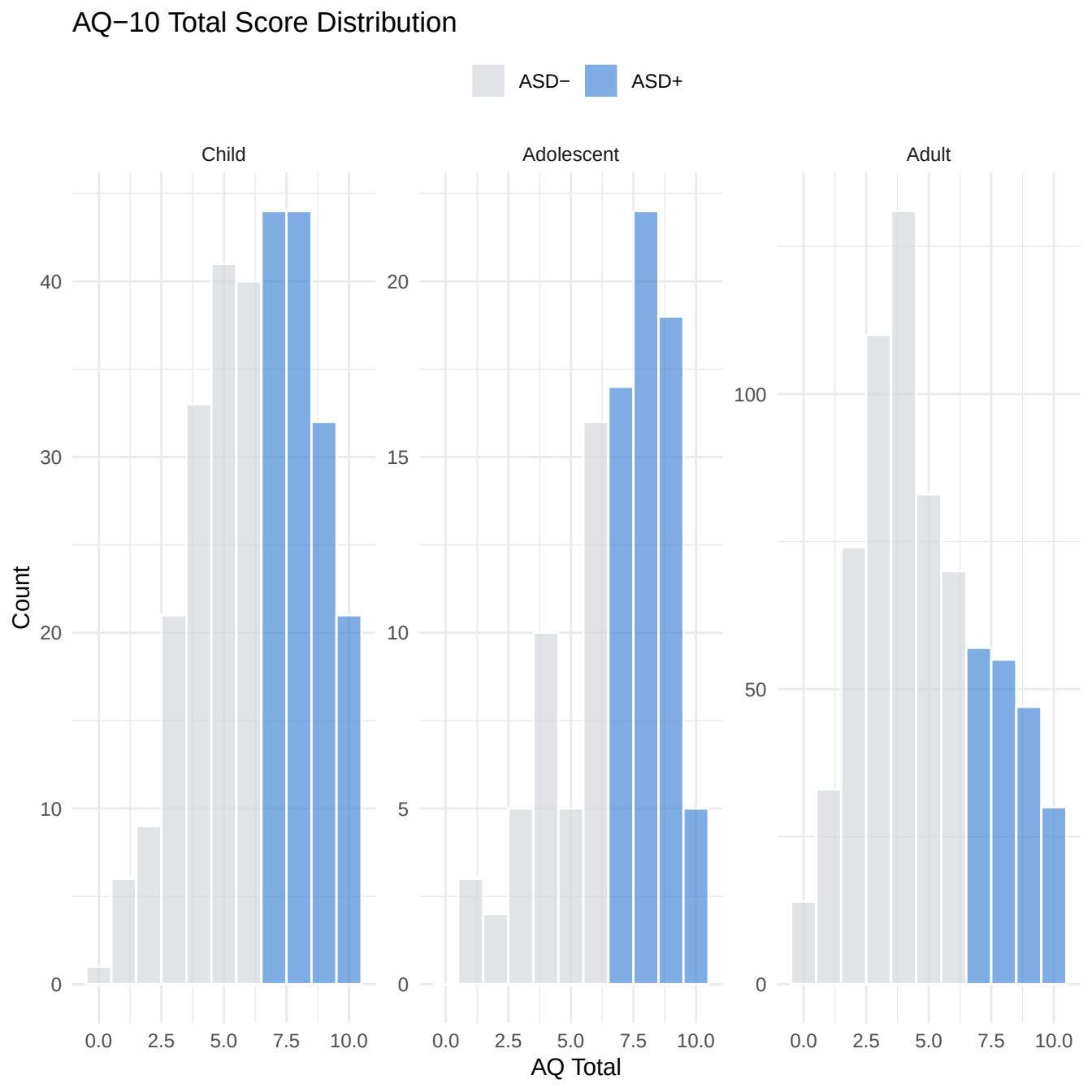
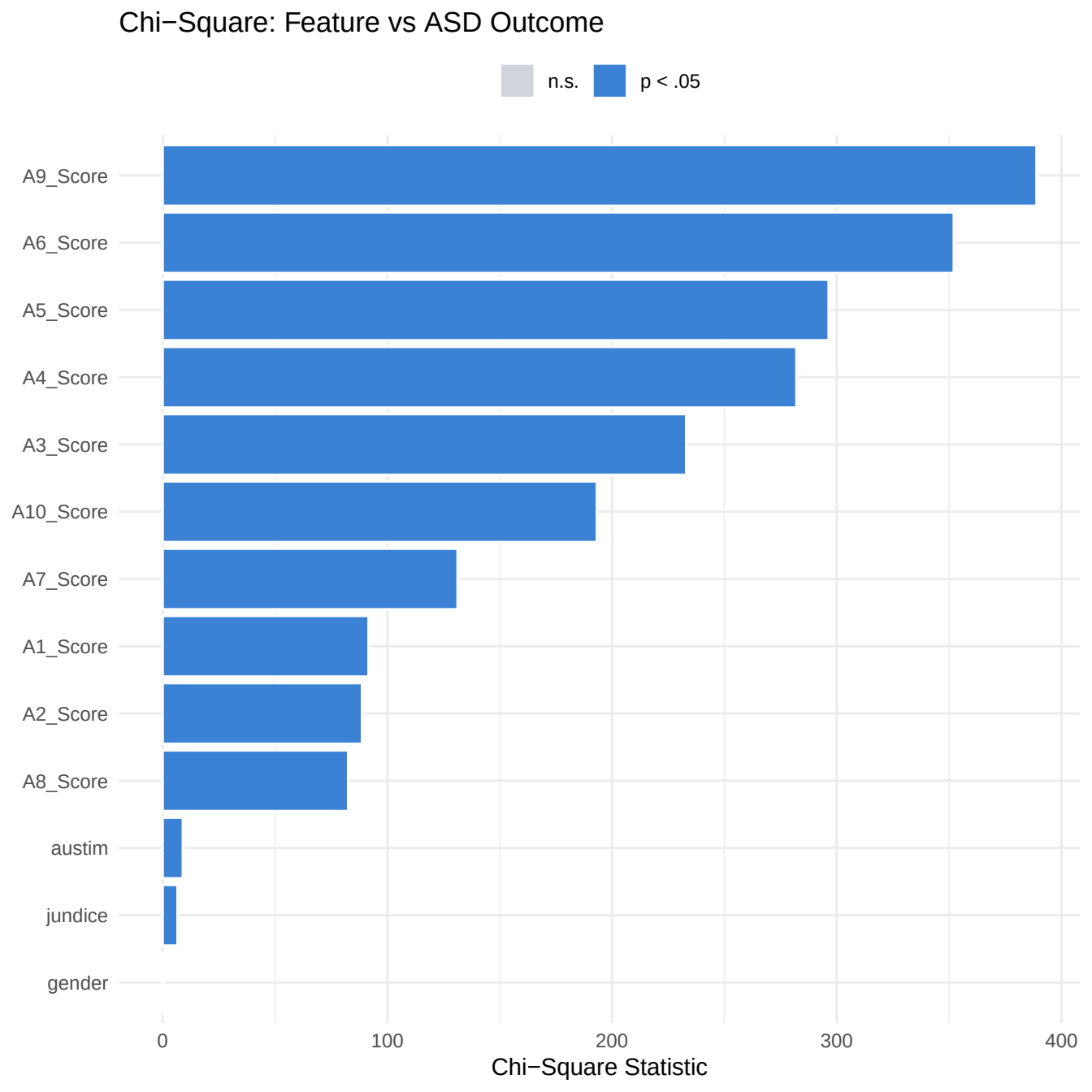
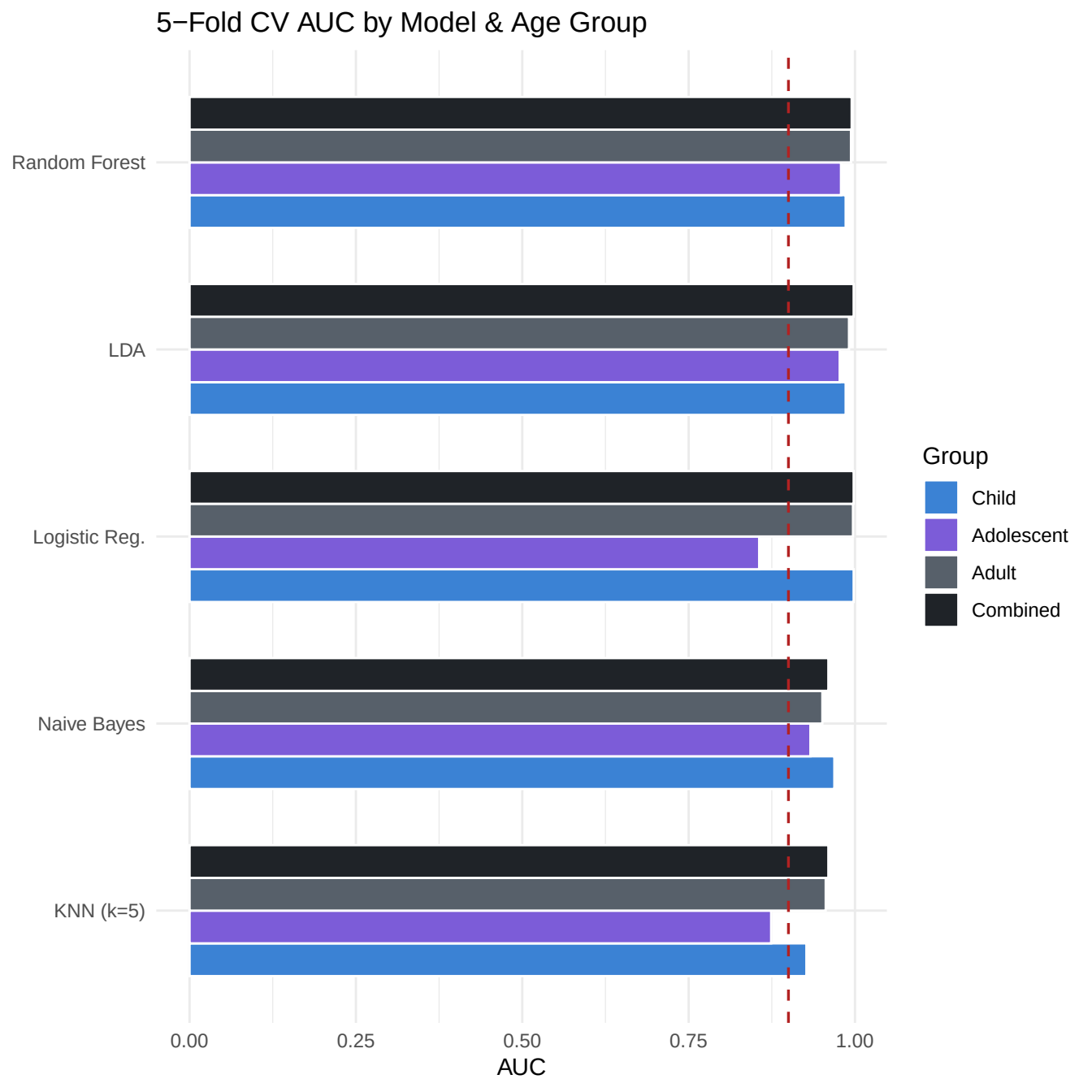
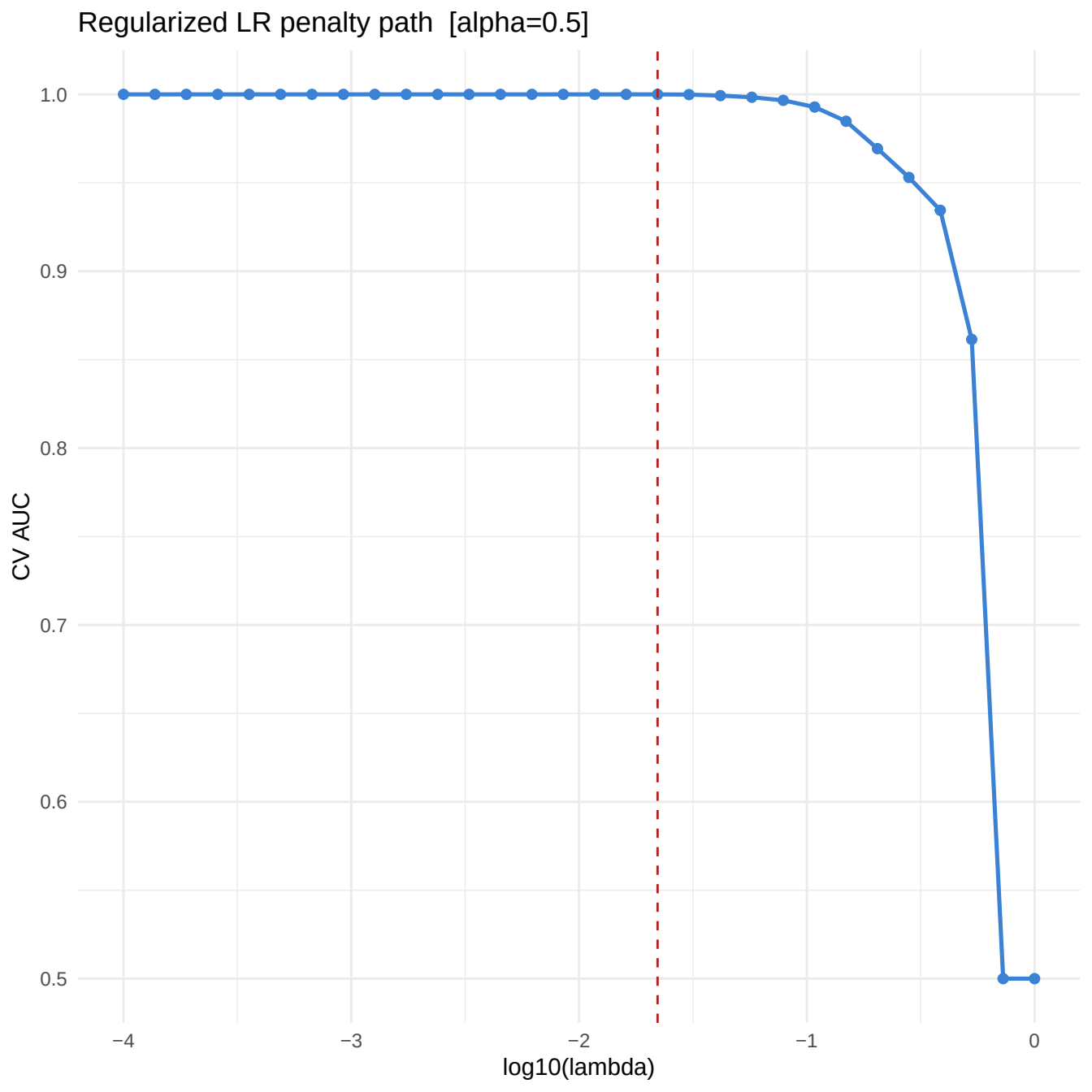
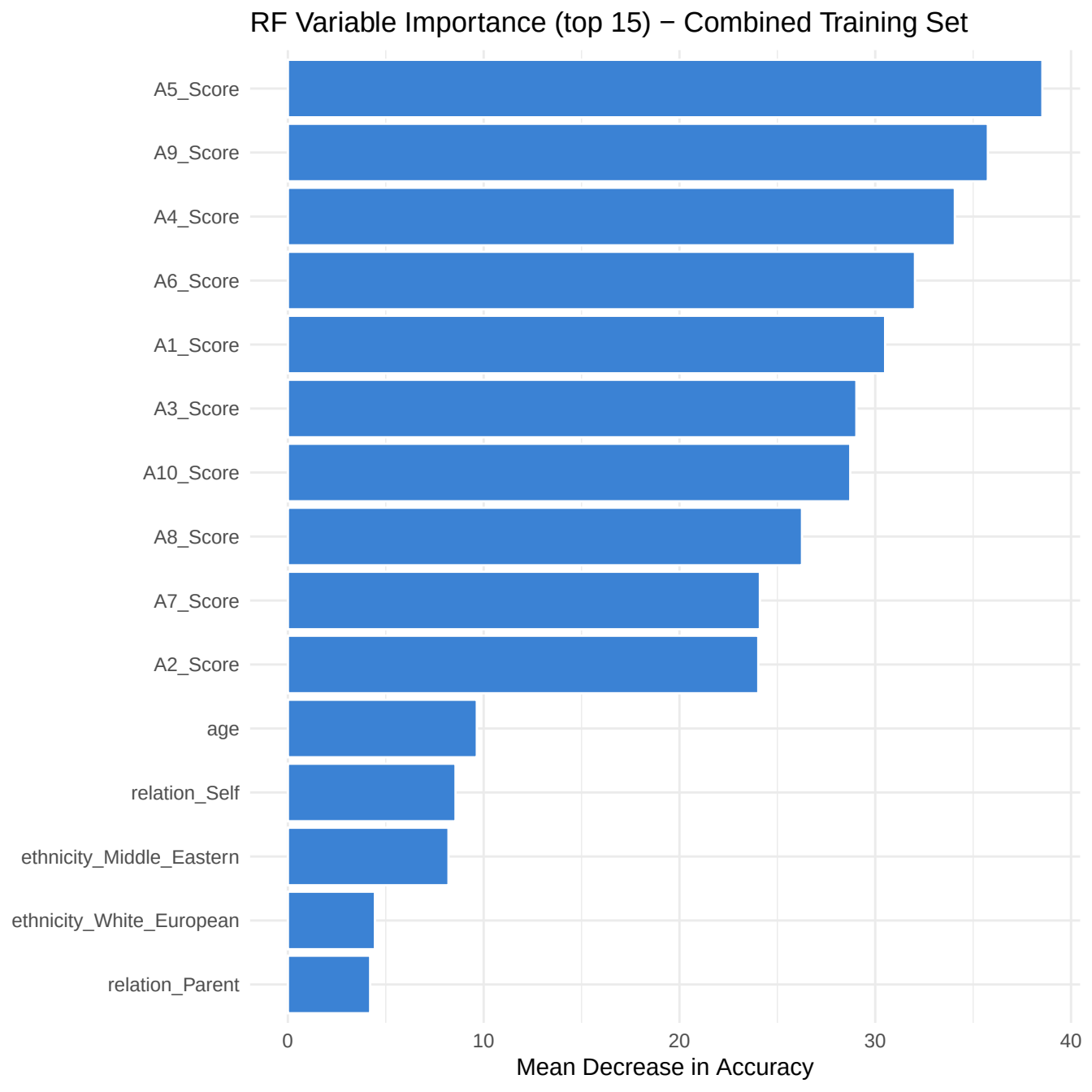

Class Balance by Age Group
ASD−
ASD+
% of group
60
40
20
0
Child
Adolescent
Adult
Combined

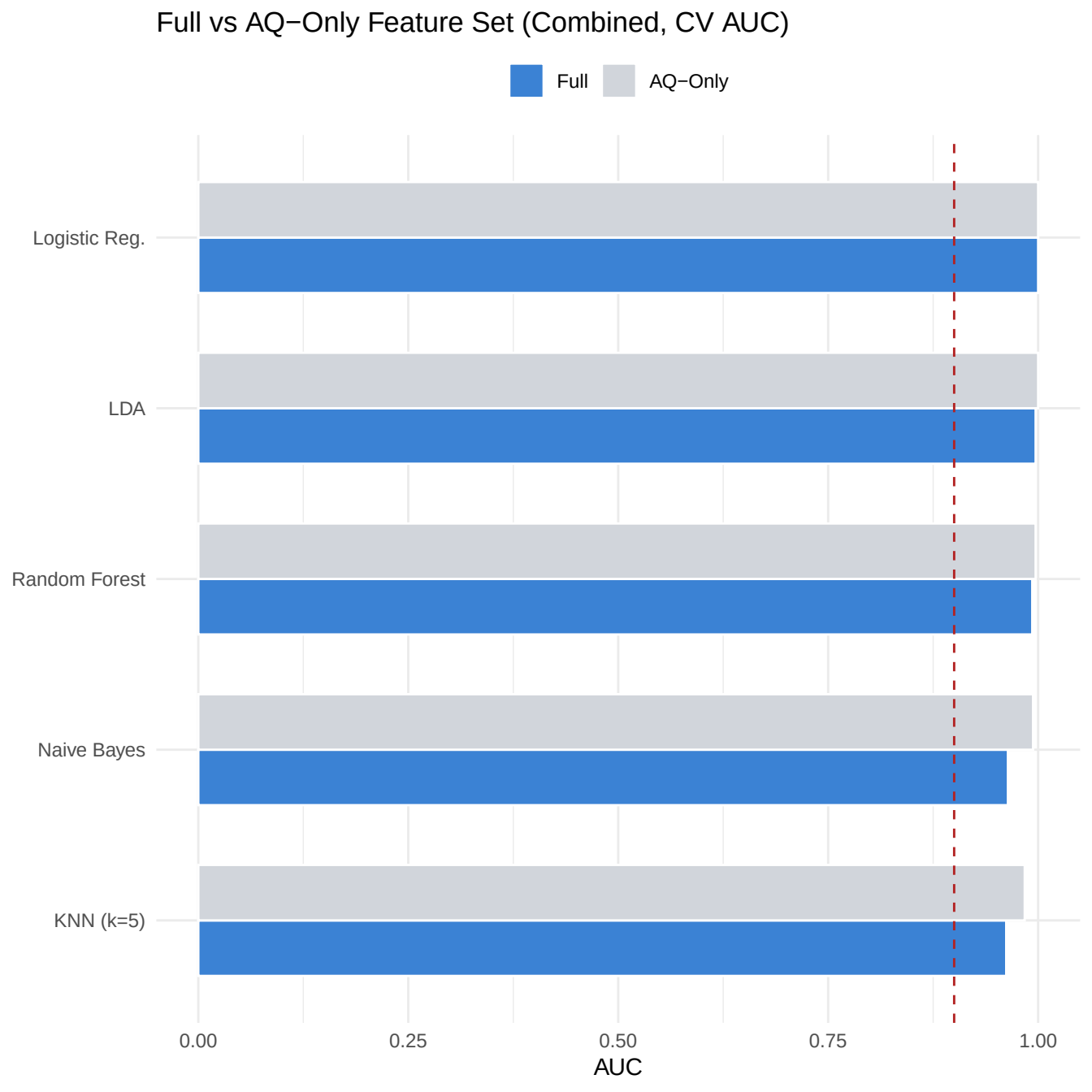

AQ−10 Total Score Distribution
ASD−
ASD+
Child
Adolescent
Adult
Count
AQ Total
0
10
20
30
40
5
15
50
100
0.0
2.5
5.0
7.5
10.0


Chi−Square: Feature vs ASD Outcome
n.s.
p < .05
A9_Score
A6_Score
A5_Score
A4_Score
A3_Score
A10_Score
A7_Score
A1_Score
A2_Score
A8_Score
austim
jundice
gender
0
100
200
300
400
Chi−Square Statistic


5−Fold CV AUC by Model & Age Group
Random Forest
LDA
Logistic Reg.
Naive Bayes
KNN (k=5)
0.00
0.25
0.50
0.75
1.00
AUC
Group
Child
Adolescent
Adult
Combined


Regularized LR penalty path [alpha=0.5]
CV AUC
1.0
0.9
0.8
0.7
0.6
0.5
−4
−3
−2
−1
0
log10(lambda)


RF Variable Importance (top 15) − Combined Training Set
A5_Score
A9_Score
A4_Score
A6_Score
A1_Score
A3_Score
A10_Score
A8_Score
A7_Score
A2_Score
age
relation_Self
ethnicity_Middle_Eastern
ethnicity_White_European
relation_Parent
0
10
20
30
40
Mean Decrease in Accuracy


Full vs AQ−Only Feature Set (Combined, CV AUC)
Full
AQ−Only
Logistic Reg.
LDA
Random Forest
Naive Bayes
KNN (k=5)
0.00
0.25
0.50
0.75
1.00
AUC